Sachchit Balamurugan

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PROFESSIONAL SUMMARY

Bioengineering student at UC Berkeley (Regents' and Chancellors' Scholar) with hands-on experience in computational biology, bioinformatics, AI/ML research, and applied data science. Published cancer detection research presented at the University of Tokyo; conducted molecular docking and quantum chemistry research at SMU; contributed to production AWS infrastructure at State Farm. Certiport-certified in AI, Python, Java, and JavaScript.

EDUCATION

University of California, Berkeley
B.S. Bioengineering. Regents' and Chancellors' Scholar. *May 2030*
Relevant Coursework: AP Computer Science A, AP Computer Science Principles, AP Statistics, AP Calculus AB, AP Biology, AP Chemistry, Network+, Operating System Security, Fundamentals of Information Security, Cloud Computing, Calculus II

Texas Online Preparatory School TX
High School Diploma. Rank: 1 / 514. GPA: 5.38. *May 2026*

EXPERIENCE

Co-Founder & Chief Technology Officer July 2026 – Present
FschoolAI (fschoolai.com)
- Co-founded an AI tutoring startup that plugs into Canvas and other school LMS platforms and teaches from each student's own course materials, backed by a grade guarantee that refunds up to $2,000 if a student fails
- Lead engineering and product: built the React/TypeScript app, serverless API, and Postgres RAG pipeline, plus a cited LLM tutor with voice, live study rooms with an AI whiteboard, and a Chrome extension that syncs school LMS data
- Grew to 130K+ users, a 5K+ student waitlist, and 600+ beta testers across 90+ schools; 5M+ social media views in 30 days
- Interviewed for PearX, Pear VC's startup accelerator

Research Intern August 2023 – Present
Southern Methodist University
- Conducted quantum chemistry calculations on 2,4,6-Triaminopyrimidine (TAP) to evaluate its potential as an RNA ancestor, performing optimizations, frequency analyses, and state determinations under UV radiation and primordial Earth conditions
- Collaborated with the SMU team to study malathion's inhibition of human acetylcholinesterase using molecular docking; proposed a two-step inhibition mechanism based on computational results

NLP Research Intern June 2024 – August 2024
University of Texas at Dallas
- Conducted applied NLP research on part-of-speech tagging, word sense disambiguation, co-reference resolution, and document classification across news, social media, and medical records datasets
- Collaborated with faculty to strengthen interdisciplinary applications of AI in linguistics and healthcare

Software/Data Infrastructure Engineer Intern June 2026 – Present
State Farm
- Building an AI infrastructure pipeline that converts test case data and uses LLMs to generate descriptions for State Farm's internal Test Case AI model
- Invited back as a returning intern after Summer 2025 engagement; continuing infrastructure and AI/ML development work across summer and fall

Software/Data Infrastructure Engineer Intern June 2025 – August 2025
State Farm
- Selected from 9,000+ applicants (2% acceptance rate); built AWS infrastructure with OpenTofu and Python for the SpaceX team, automating secure token access and AI-powered merge request notifications
- Contributed to a hackathon project developing an AI-driven employee dictionary for rapid term search

Air Quality Research Intern May 2024 – June 2026
University of Texas at Dallas
- Building an interactive web platform (Vue.js) to visualize real-time particulate matter and wind data for academic and public use in air quality monitoring

Research Intern June 2024 – September 2024
Duke University
- Analyzed correlations between U.S. household income and Universal Tennis Ratings (UTR) for youth players aged 10-18; applied regression and hypothesis testing to identify socioeconomic disparities
- Published findings in the Journal of Research High School

AI Software Engineer Intern August 2024 – June 2026
Gigaplex Academy
- Built Python scripts, algorithms, and AI chatbots to recommend FBLA/CTSO competitions tailored to student interests and personalities
- Featured on Gigaplex Academy's website for cancer research contributions; improved its chatbot to better handle program inquiries

Chief Technology Officer June 2024 – June 2026
EcoBox LLC
- Built EcoWise, a sustainability app published on Google Play; directed the team in designing an AI-powered carbon tracking app with gamified features and sold 500+ EcoBox workbooks to schools

Research Intern October 2024 – June 2025
NASA (Texas High School Aerospace Scholars & Moonshot Internship)
- Selected for NASA's HAS program; completed a 7-month aerospace curriculum and advanced to the Moonshot Summer Virtual Intensive Internship with 8-hour daily sessions and engineering challenges
- Applied systems engineering, CAD modeling, orbital mechanics, and propulsion analysis to design solutions for Artemis mission challenges

PROJECTS

Independent Research: Cancer Detection with Machine Learning https://docs.google.com/presentation/d/11sLJhjsHqOyLSVbSmTSELfLPORNEjzzRuKwtCC10Zgk/edit?usp=sharing May 2023 – Present
- Pulled transcriptome profiling data directly from the TCGA database using R for bioinformatics preprocessing; combined with public MRI images to train deep learning models (ResNet50, MobileNetV2, VGG, ConvNeXtTiny, and custom CNN) for early detection of Adrenocortical Carcinoma and six secondary tumors
- Identified two novel biomarkers across 7 cancer types using statistical modeling and data analysis; visualized findings with Matplotlib and pandas
- Selected as one of 27 students worldwide to present at the International Youth Research Conference at the University of Tokyo; published in the IYRC Peer-Reviewed Journal (DOI: 10.34614/JIYRC2025II23)
- Featured in Forbes, D Magazine, Dallas Innovates, and Spectrum News (combined reach: 5M+ viewers)

Soul Sync Mental Health App June 2023 – June 2025
- Created a mental health app for OCD and depression featuring an AI chatbot, ERP therapy tools, and mood tracking; partnered with OCD Texas to refine features and distributed to community members
- Recognized as a RISE Global Challenge Finalist (Top ~2%)

Youth Opportunities in Tech Innovation (YOTI) https://yotiweb.com April 2024 – June 2026
Founder & CEO
- Founded a nonprofit teaching coding and app/web development to 100+ global students; secured $600 in grants; hosted AI/ML workshops with 85+ participants
- Built free websites for organizations including Project Surya, which now reaches 4 million viewers

ACTIVITIES & LEADERSHIP

FIRST Robotics Competition (FRC), Mercenary Robotics April 2022 – June 2026
Software Engineer / Design Lead
- Led design team on FRC competition robots, developing CAD systems, Swerve drive, and motor-powered shooter; built a scouting app and AI tools for data collection and alliance selection at State and World competitions

Business Professionals of America (BPA) August 2023 – June 2026
Chapter Vice President
- Led chapter initiatives and mentored peers toward Torch Awards; first national winner from school in Mobile Application competition; interned at State and National Leadership Conferences (2024, 2025)

Future Business Leaders of America (FBLA) August 2023 – June 2026
Texas Area 9 State VP & Chapter VP
- Elected Texas FBLA Area 9 State VP (2025-26); created FBLA's first-ever gamified service challenge; built an AI chatbot to match students with FBLA events; planned the 2026 State Leadership Conference

Civics Unplugged Fellowships & Innovation Academies November 2023 – August 2026
Fellow & Workshop Presenter

- Selected for three fellowships (~10% acceptance rate); at UCLA (2024), built an algorithm creating a priority system for allocating cancer beds during pandemic conditions; presented a workshop at SXSW EDU 2026 on cancer research and YOTI

AWARDS & HONORS

- NASA Space App Challenge: Global Nominee & Global People's Choice Award, Climate Change Challenge (2024); Global Nominee (2025) -- Top 940 of 15,000 teams
- RISE Global Challenge Finalist (Top ~2%, 2023): Recognized for AI cancer prediction model and Soul Sync mental health app
- Business Professionals of America: National 1st Place, Mobile Application (2024-25); National 7th Place, Computer Modeling (2023-24); Statesman Torch Award
- FIRST Robotics (FRC) 2025: Ranked 27th/3,690 at Worlds (21st USA, 3rd Texas); Texas State Champion; District Champion (Amarillo & Belton, TX)
- Destination Imagination 2023: World Champion, Digital Open Challenge; Global Finals 1st Place, Service & Instant Challenges; Magellan Award
- International Science and Engineering Fair State Qualifier 2024: 3rd place, Fort Worth Regional in Computer Science; advanced to Texas A&M State Fair
- United Nations Association: National Ambassador Award; Zero Hunger & Zero Waste Awards (Gold & Green); 1,200+ service hours
- President's Volunteer Service Award: Gold Medal (2024-2025)
- Boy Scouts of America: Eagle Scout; Silver, Gold, and Bronze Eagle Palms; 45 merit badges; Order of the Arrow
- College Board: AP Scholar with Distinction (2025); National Recognition Program (2025)

CERTIFICATIONS

- Certiport IT Specialist: Artificial Intelligence, Python, Java, JavaScript, HTML/CSS
- Microsoft Office Specialist: Word Associate (Office 2019)
- BPA Student Certification Series: Level One

SKILLS

Computational Biology & Bioinformatics: Computational Biology, Bioinformatics, Statistical Modeling, Data Analysis, R, pandas, Matplotlib, Machine Learning, Deep Learning, Artificial Intelligence, Computer Vision, NLP, TensorFlow, scikit-learn, Pretrained Neural Networks, Prompt Engineering
Programming & Development: Python, R, Java, JavaScript, Dart, React JS, Flutter, Firebase, HTML/CSS, Vue.js, Flask
Cloud & Infrastructure: AWS, OpenTofu, Cloud Computing, Cybersecurity
Software & Platforms: Android Studio, Xcode, Arduino, CAD, Blender
Robotics: EV3, Spike Prime, Arduino-based systems, VEX Robotics